



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 2

2022 Year 12 Course Assessment Task 3

Thursday June 9th, 2022

General instructions

- Working time – 1 hour.
(plus 5 minutes reading time)
- Write your answers in the space provided.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.
- At the conclusion of the examination, bundle any additional writing booklets used in the correct order within this paper and hand to examination supervisors.
- A NESA Reference Sheet is provided.

Class (please ✓)

- ☐ 12MXX.1 – Ms Ham
- ☐ 12MXX.2 – Mr Sekaran
- ☐ 12MXX.3 – Mr Ho

NESA STUDENT #: # BOOKLETS USED:

Marker's use only.

QUESTION	1	2	3	Total	%
MARKS	$\overline{17}$	$\overline{14}$	$\overline{9}$	$\overline{40}$	

Glossary

- $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3\}$ – set of all integers.
- $\mathbb{Z}^+$ – all positive integers (excludes zero)
- $\mathbb{R}$ – set of all real numbers

Question 1 (17 Marks)**Marks**

- (a) i. Using partial fractions, or otherwise, find **3**

$$\int_1^3 \frac{1}{x(4-x)} dx$$

- ii. Let $I = \int_1^3 \frac{\cos^2\left(\frac{\pi}{8}x\right)}{x(4-x)} dx$. **2**

Use the substitution $u = 4 - x$ to show that

$$I = \int_1^3 \frac{\sin^2\left(\frac{\pi}{8}u\right)}{u(4-u)} du$$

- iii. Hence, find the value of I . **1**

- (b) Find **3**

$$\int \sqrt{\frac{3-x}{x}} dx$$

- (c) Use a suitable substitution to find **4**

$$\int_2^4 \frac{dx}{x^2\sqrt{x^2-4}}$$

- (d) Let $I = \int_{(k-1)\pi}^{k\pi} e^x \sin x dx$, where $k \in \mathbb{Z}$. Show that **4**

$$I = \frac{(-1)^{k-1} e^{k\pi} (e^\pi + 1)}{2e^\pi}$$

Examination continues overleaf...

Question 2 (14 Marks)**Marks**

- (a) Let $\underline{a} = 2\underline{i} - 3\underline{j} + \underline{k}$ and $\underline{b} = \underline{i} + m\underline{j} - \underline{k}$, where m is a positive integer.

The projection of $\underline{a}$ onto $\underline{b}$ is $-\frac{11}{18}(\underline{i} + m\underline{j} - \underline{k})$.

- i. Find the value of m .

2

- ii. Find the component of $\underline{a}$ that is perpendicular to $\underline{b}$.

1

- (b) The line ℓ_1 has vector equation $\underline{r}_1 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda_1 \begin{pmatrix} 2 \\ 2 \\ -3 \end{pmatrix}$ and the line ℓ_2 has vector

equation $\underline{r}_2 = \begin{pmatrix} 4 \\ -2 \\ 9 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$.

- i. Show that the distance d between a point on ℓ_1 and a point on ℓ_2 is given by

1

$$d = \sqrt{(3 - 2\lambda_1 + \lambda_2)^2 + (-2 - 2\lambda_1 + 2\lambda_2)^2 + (7 + 3\lambda_1 - 2\lambda_2)^2}$$

It is given that

$$\begin{aligned} (3 - 2\lambda_1 + \lambda_2)^2 + (-2 - 2\lambda_1 + 2\lambda_2)^2 + (7 + 3\lambda_1 - 2\lambda_2)^2 \\ = (3\lambda_2 - 4\lambda_1 - 5)^2 + (\lambda_1 - 1)^2 + 36 \end{aligned}$$

(DO NOT PROVE THIS)

- ii. Hence, or otherwise, determine the minimum distance between ℓ_1 and ℓ_2 .

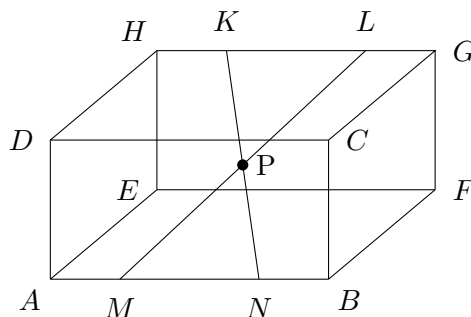
1

- iii. Find the coordinates of the points on the two lines that are the minimum distance apart.

2

- (c) The diagram shows a rectangular prism whose vertices are A, B, C, D, E, F, G and H . The points M and N lie on AB such that $AM : MN : NB = 1 : 2 : 1$. The points K and L lie on HG such that $HK : KL : LG = 1 : 2 : 1$. P is the intersection of KN and ML .

Let $\overrightarrow{AB} = \underline{a}$, $\overrightarrow{AE} = \underline{e}$ and $\overrightarrow{AD} = \underline{d}$.



- i. Show that $\overrightarrow{ML} = \frac{1}{2}\underline{a} + \underline{d} + \underline{e}$

2

- ii. Find a similar expression for $\overrightarrow{NK}$ in terms of $\underline{a}$, $\underline{d}$ and $\underline{e}$

1

- iii. By using $\overrightarrow{AP}$, prove that KN and ML bisect each other at P .

4

Question 3 (9 Marks)**Marks**

(a) It is given that $a, b, c \in \mathbb{R}^+$ and $a + b + c = 1$.

i. Prove $\frac{a}{b} + \frac{b}{a} \geq 2$. **1**

ii. Prove $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$. **2**

iii. Hence, or otherwise, prove $(1 - a)(1 - b)(1 - c) \geq 8abc$. **2**

(b) A function is defined to be bounded if $\exists M \in \mathbb{R}$ such that $|f(x)| \leq M$.

It is given that $g(x)$ and $h(x)$ are continuous and bounded functions.

i. By considering $\int_0^a [\lambda g(x) + h(x)]^2 dx$ for $a > 0$ and $\lambda \in \mathbb{R}$ as a quadratic function in λ , show that **2**

$$\left[\int_0^a g(x)h(x) dx \right]^2 \leq \int_0^a [g(x)]^2 dx \times \int_0^a [h(x)]^2 dx$$

ii. Hence show that **1**

$$\left[\int_0^1 g(x) dx \right]^2 \leq \int_0^1 [g(x)]^2 dx$$

iii. Deduce that **1**

$$\left[\int_0^1 g(x) dx \right]^4 \leq \int_0^1 [g(x)]^4 dx$$

End of paper.

Sample Band E4 Responses

Question 1 (Sekaran and Ham(d))

(a) i. (3 marks)

✓ [1] for A and B

✓ [1] for correct integration

✓ [1] for final answer

$$\text{Let } \frac{1}{x(4-x)} = \frac{A}{x} + \frac{B}{4-x}$$

$$1 = A(4-x) + Bx$$

$$A = \frac{1}{4}, B = \frac{1}{4}$$

$$\begin{aligned} \int_1^3 \frac{1}{x(4-x)} dx &= \frac{1}{4} \int_1^3 \frac{1}{x} + \frac{1}{4-x} dx \\ &= \frac{1}{4} [\ln|x| - \ln|4-x|]_1^3 \\ &= \frac{1}{4} (\ln 3 - \ln 1 - (\ln 1 - \ln 3)) \\ &= \frac{\ln 3}{2} \end{aligned}$$

ii. (2 marks)

✓ [1] for $du = -dx$ and the new limits

✓ [1] for final result

Let $u = 4 - x, du = -dx$

When $x = 1, u = 3$ and $x = 3, u = 1$.

$$\begin{aligned} I &= \int_1^3 \frac{\cos^2\left(\frac{\pi}{8}x\right)}{x(4-x)} dx \\ &= - \int_3^1 \frac{\cos^2\left[\frac{\pi}{8}(4-u)\right]}{(4-u)u} du \\ &= \int_1^3 \frac{\cos^2\left(\frac{\pi}{2} - \frac{\pi}{8}u\right)}{u(4-u)} du \\ &= \int_1^3 \frac{\sin^2\frac{\pi}{8}u}{u(4-u)} du \end{aligned}$$

iii. (1 mark)

$$\text{From (ii), } I = \int_1^3 \frac{\sin^2\frac{\pi}{8}x}{x(4-x)} dx$$

$$\begin{aligned} 2I &= \int_1^3 \frac{\cos^2\frac{\pi}{8}x}{x(4-x)} + \frac{\sin^2\frac{\pi}{8}x}{x(4-x)} dx \\ &= \int_1^3 \frac{1}{x(4-x)} dx \\ &= \frac{\ln 3}{2} \dots \text{from part (i)} \\ \therefore I &= \frac{\ln 3}{4} \end{aligned}$$

(b) (3 marks)

- ✓ [1] for correctly completing the square.
- ✓ [1] for substantial progress to final answer
- ✓ [1] for final answer

$$\begin{aligned}
 \int \sqrt{\frac{3-x}{x}} dx &= \int \frac{3-x}{x\sqrt{3-x}} dx \\
 &= \int \frac{3-x}{\sqrt{-x^2+3x}} dx \\
 &= \int \frac{3-x}{\sqrt{-(x-\frac{3}{2})^2 + (\frac{3}{2})^2}} dx \\
 &= \int \frac{\frac{1}{2}(3-2x) + \frac{3}{2}}{\sqrt{-(x-\frac{3}{2})^2 + (\frac{3}{2})^2}} dx \\
 &= \int \frac{\frac{1}{2}(3-2x) + \frac{3}{2}}{\sqrt{-(x-\frac{3}{2})^2 + (\frac{3}{2})^2}} dx \\
 &= \frac{1}{2} \int \frac{3-2x}{\sqrt{3x-x^2}} dx + \frac{3}{2} \int \frac{1}{\sqrt{(\frac{3}{2})^2 - (x-\frac{3}{2})^2}} dx \\
 &= \frac{1}{2} \sqrt{3x-x^2} \times 2 + \frac{3}{2} \sin^{-1} \left(\frac{x-\frac{3}{2}}{\frac{3}{2}} \right) + C \\
 &= \sqrt{3x-x^2} + \frac{3}{2} \sin^{-1} \left(\frac{2x-3}{3} \right) + C
 \end{aligned}$$

(c) (4 marks)

- ✓ [1] for transforming the differential.
- ✓ [1] for transforming both limits.
- ✓ [1] for transforming integrand to an integrable form.
- ✓ [1] for final answer.

$$\begin{aligned}
 x &= 2 \sec \theta \\
 \therefore dx &= 2 \sec \theta \tan \theta d\theta
 \end{aligned}$$

Transforming the limits,

$$\begin{aligned}
 x = 4 \quad \theta &= \frac{\pi}{3} \\
 x = 2 \quad \theta &= 0
 \end{aligned}$$

$$\begin{aligned}
& \int_0^{\frac{\pi}{3}} \frac{1}{4 \sec^2 \theta \sqrt{4 \sec^2 \theta - 4}} 2 \sec \theta \tan \theta d\theta \\
&= \frac{1}{2} \int_0^{\frac{\pi}{3}} \frac{\tan \theta}{\sec \theta \times 2 \tan \theta} d\theta \\
&= \frac{1}{4} \int_0^{\frac{\pi}{3}} \frac{1}{\sec \theta} d\theta \\
&= \frac{1}{4} \int_0^{\frac{\pi}{3}} \cos \theta d\theta &= \frac{1}{4} \left[\sin \theta \right]_0^{\frac{\pi}{3}} \\
&= \frac{1}{4} \times \frac{\sqrt{3}}{2} \\
&= \frac{\sqrt{3}}{8}
\end{aligned}$$

(d) (4 marks)

- ✓ [1] for one application of integrating by parts
- ✓ [1] for another application of integrating by parts
- ✓ [1] for obtaining $2I$
- ✓ [1] for evaluating $\cos(k\pi)$ and $\cos((k-1)\pi)$ and getting the final result

$$\begin{aligned}
I &= \int_{(k-1)\pi}^{k\pi} e^x \sin x dx \\
&\left| \begin{array}{ll} u = \sin x & v = e^x \\ du = \cos x & dv = e^x \end{array} \right. \\
I &= \left[e^x \sin x \right]_{(k-1)\pi}^{k\pi} - \int_{(k-1)\pi}^{k\pi} e^x \cos x dx \\
&\left| \begin{array}{ll} u = \cos x & v = e^x \\ du = -\sin x & dv = e^x \end{array} \right. \\
&= 0 - \left(\left[e^x \cos x \right]_{(k-1)\pi}^{k\pi} + \int_{(k-1)\pi}^{k\pi} e^x \sin x dx \right) \\
\therefore 2I &= -e^{k\pi} \cos(k\pi) + e^{(k-1)\pi} \cos(k-1)\pi \\
&= -e^{k\pi} (-1)^k + e^{(k-1)\pi} (-1)^{k-1} \\
&= e^{k\pi} (-1)^{k+1} + e^{(k-1)\pi} (-1)^{k-1} \\
&= e^{(k-1)\pi} (-1)^{k-1} [(-1)^2 e^\pi + 1] \\
\therefore I &= \frac{(-1)^{k-1} e^{k\pi} (e^\pi + 1)}{2e^\pi}
\end{aligned}$$

Question 2 (Ho)

(a) i. (2 marks)

✓ [1] for $\text{proj}_{\underline{b}} \underline{a}$

✓ [1] for final answer

$$\begin{aligned}\text{proj}_{\underline{b}} \underline{a} &= \frac{2-3m-1}{1^2+m^2+1^2} (\underline{i} + m\underline{j} - \underline{k}) \\ &= \frac{1-3m}{2+m^2} (\underline{i} + m\underline{j} - \underline{k})\end{aligned}$$

$$\begin{aligned}\therefore \frac{1-3m}{2+m^2} &= -\frac{11}{18} \\ 18-54m &= -11m^2-22 \\ 11m^2-54m+40 &= 0 \\ (11m-10)(m-4) &= 0 \\ \therefore m &= \frac{10}{11} \text{ or } m = 4\end{aligned}$$

But $\because m \in \mathbb{Z}^+$, $m = 4$.

ii. (1 mark)

$$2\underline{i} - 3\underline{j} + \underline{k} - \left(-\frac{11}{18}\right) (\underline{i} + 4\underline{j} - \underline{k}) = \frac{47}{18}\underline{i} + \frac{-5}{9}\underline{j} - \frac{7}{18}\underline{k}$$

(b) i. (1 mark)

$$\begin{aligned}d &= \left| \begin{pmatrix} 4+\lambda_2 \\ -2+2\lambda_2 \\ 9-2\lambda_2 \end{pmatrix} - \begin{pmatrix} 1+2\lambda_1 \\ 2\lambda_1 \\ 2-3\lambda_1 \end{pmatrix} \right| \\ &= \left| \begin{pmatrix} 3-2\lambda_1+\lambda_2 \\ -2-2\lambda_1+2\lambda_2 \\ 7+3\lambda_1-2\lambda_2 \end{pmatrix} \right| \\ &= \sqrt{(3-2\lambda_1+\lambda_2)^2 + (-2-2\lambda_1+2\lambda_2)^2 + (7+3\lambda_1-2\lambda_2)^2}\end{aligned}$$

ii. (1 mark) Minimum $d = \sqrt{36} = 6$

iii. (2 marks)

✓ [1] for λ_1 and λ_2 .

✓ [1] for final answer

Minimum d when $3\lambda_2 - 4\lambda_1 - 5 = 0$ and $\lambda_1 - 1 = 0$

$$\therefore \lambda_1 = 1 \text{ and } \lambda_2 = 3$$

When $\lambda_1 = 1$,

$$\underline{r}_1 = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix}$$

When $\lambda_2 = 3$,

$$\underline{r}_2 = \begin{pmatrix} 4 \\ -2 \\ 9 \end{pmatrix} + \begin{pmatrix} 3 \\ 6 \\ -6 \end{pmatrix} = \begin{pmatrix} 7 \\ 4 \\ 3 \end{pmatrix}$$

$\therefore$ The coordinates are $(3, 2, -1)$ and $(7, 4, 3)$.

(c) i. (2 marks)

✓ [1] for (1) or equivalent

✓ [1] for final result

$$\begin{aligned} \overrightarrow{ML} &= \overrightarrow{MA} + \overrightarrow{AD} + \overrightarrow{DH} + \overrightarrow{HL} \dots (1) \\ &= -\frac{1}{4}\underline{a} + \underline{d} + \underline{e} + \frac{3}{4}\underline{a} \\ &= \frac{1}{2}\underline{a} + \underline{d} + \underline{e} \end{aligned}$$

ii. (1 mark)

$$\begin{aligned} \overrightarrow{NK} &= \overrightarrow{NB} + \overrightarrow{BF} + \overrightarrow{FG} + \overrightarrow{GK} \\ &= \frac{1}{4}\underline{a} + \underline{e} + \underline{d} - \frac{3}{4}\underline{a} \\ &= -\frac{1}{2}\underline{a} + \underline{d} + \underline{e} \end{aligned}$$

iii. (4 marks)

✓ [1] for one of two expressions of $\overrightarrow{AP}$

✓ [1] for the other expression of $\overrightarrow{AP}$

✓ [1] for substantial progress to find λ and μ

✓ [1] for final result

$$\begin{aligned}
\overrightarrow{AP} &= \overrightarrow{AM} + \overrightarrow{MP} \\
&= \overrightarrow{AM} + \lambda \overrightarrow{ML} \quad \dots \text{where } \lambda \in \mathbb{R} \\
&= \frac{1}{4}\mathbf{a} + \lambda \left(\frac{1}{2}\mathbf{a} + \mathbf{d} + \mathbf{e} \right) \\
\overrightarrow{AP} &= \overrightarrow{AN} + \overrightarrow{NP} \\
&= \overrightarrow{AN} + \mu \overrightarrow{NK} \quad \dots \text{where } \mu \in \mathbb{R} \\
&= \frac{3}{4}\mathbf{a} + \mu \left(-\frac{1}{2}\mathbf{a} + \mathbf{d} + \mathbf{e} \right)
\end{aligned}$$

$$\begin{aligned}
\therefore \frac{1}{4}\mathbf{a} + \lambda \left(\frac{1}{2}\mathbf{a} + \mathbf{d} + \mathbf{e} \right) &= \frac{3}{4}\mathbf{a} + \mu \left(-\frac{1}{2}\mathbf{a} + \mathbf{d} + \mathbf{e} \right) \\
\frac{1}{2}(1 - \mu - \lambda)\mathbf{a} + (\mu - \lambda)\mathbf{d} + (\mu - \lambda)\mathbf{e} &= \mathbf{0}
\end{aligned}$$

$$\begin{aligned}
\therefore \mu &= \lambda \text{ and } 1 - \mu - \lambda = 0 \\
1 - 2\lambda &= 0 \\
\lambda = \mu &= \frac{1}{2}
\end{aligned}$$

Hence KN and ML bisect each other at P .

Question 3 (Ham)

(a) (1 mark)

$$\begin{aligned}\left(\sqrt{\frac{a}{b}} - \sqrt{\frac{b}{a}}\right)^2 &\geq 0 \\ \frac{a}{b} - 2 + \frac{b}{a} &\geq 0 \\ \therefore \frac{a}{b} + \frac{b}{a} &\geq 2\end{aligned}$$

(b) (2 marks)

- ✓ [1] for (1).
 ✓ [1] for final result.

$$\begin{aligned}\frac{1}{a} + \frac{1}{b} + \frac{1}{c} &= (a+b+c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \quad \dots (1) \\ &= 1 + 1 + 1 + \frac{a}{b} + \frac{b}{a} + \frac{a}{c} + \frac{c}{a} + \frac{b}{c} + \frac{c}{b} \\ &\geq 3 + 2 + 2 + 2 \\ &= 9\end{aligned}$$

(c) (2 marks)

- ✓ [1] for (*).
 ✓ [1] for final result.

$$\begin{aligned}(1-a)(1-b)(1-c) &= (1-b-a+ab)(1-c) \\ &= 1-c-b+bc-a+ac+ab-abc \\ &= 1-(a+b+c)+ab+ac+bc-abc \\ &= 0 + \textcolor{violet}{ab} + \textcolor{violet}{ac} + \textcolor{violet}{bc} - abc \quad \dots (\because a+b+c=1) \quad (*) \\ &\geq \textcolor{violet}{9abc} - abc \quad \dots \left(\frac{ab+ac+bc}{abc} \geq 9, ab+ac+bc \geq 9abc \text{ from (ii)} \right) \\ &= 8abc\end{aligned}$$

OR

$$\begin{aligned}(1-a)(1-b)(1-c) &= (b+c)(c+a)(a+b) \quad \dots (\because a+b+c=1) \\ &\geq 2\sqrt{bc} \times 2\sqrt{ac} \times 2\sqrt{ab} \quad \dots \left((\sqrt{a}-\sqrt{b})^2 \geq 0, a+b \geq 2\sqrt{ab} \right) \\ &= 8\sqrt{a^2b^2c^2} \\ &= 8abc\end{aligned}$$

(d) i. (2 marks)

- ✓ [1] for expressing as a quadratic function in λ .
 ✓ [1] for final result.

$$\begin{aligned}\int_0^a [\lambda g(x) + h(x)]^2 dx &= \int_0^a \lambda^2 [g(x)]^2 + 2\lambda g(x)h(x) + [h(x)]^2 dx \\ &= \left(\int_0^a [g(x)]^2 dx \right) \lambda^2 + 2 \left(\int_0^a g(x)h(x) dx \right) \lambda + \int_0^a [h(x)]^2 dx\end{aligned}$$

Since $\int_0^a [\lambda g(x) + h(x)]^2 dx \geq 0$

$$\begin{aligned}4 \left(\int_0^a g(x)h(x) dx \right)^2 - 4 \left(\int_0^a [g(x)]^2 dx \right) \left(\int_0^a [h(x)]^2 dx \right) &\leq 0 \\ \therefore \left[\int_0^a g(x)h(x) dx \right]^2 &\leq \int_0^a [g(x)]^2 dx \times \int_0^a [h(x)]^2 dx\end{aligned}$$

ii. (1 mark)

Let $h(x) = 1$ and $a = 1$ in (i).

$$\left[\int_0^1 g(x) dx \right]^2 \leq \int_0^1 [g(x)]^2 dx \times \int_0^1 1 dx = \int_0^1 [g(x)]^2 dx$$

iii. (1 mark)

Squaring both sides in (ii),

$$\left[\int_0^1 g(x) dx \right]^4 \leq \left[\int_0^1 [g(x)]^2 dx \right]^2$$

and replace $g(x)$ with $[g(x)]^2$ in (ii), then

$$\begin{aligned}\left[\int_0^1 [g(x)]^2 dx \right]^2 &\leq \int_0^1 [g(x)]^4 dx \\ \therefore \left[\int_0^1 g(x) dx \right]^4 &\leq \left[\int_0^1 [g(x)]^2 dx \right]^2 \leq \int_0^1 [g(x)]^4 dx\end{aligned}$$